

Systematic Review and Meta-analysis

# Which Graft Is Associated With Better Outcomes in ACL Reconstruction? A Systematic Review and Network Meta-analysis of Randomized Controlled Trials

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## Abstract

**Background** In clinical practice, different surgical methods, including autografts, allografts, and artificial ligaments, have been used in reconstruction after ACL injuries. However, the available evidence is conflicting in terms of which approach has yielded better patient-reported outcomes, joint stability, and risk of graft reinjury. A network meta-analysis allows the comparison of approaches that have not been compared head-to-head in individual RCTs. Although several network

meta-analyses have been performed on this topic, they are hampered by methodological limitations.

**Questions/purposes** We performed an updated network meta-analysis of RCTs to answer the following questions: Considering different grafts applied in ACL reconstruction, which graft is superior in terms of (1) patient-reported outcomes (International Knee Documentation Committee [IKDC] objective score, Lysholm score, and Tegner score), (2) knee stability results (Lachman test, pivot-shift test, and KT-1000/2000 arthrometer side-to-side difference), and (3) risk of graft reinjury?

**Methods** We searched the PubMed, Embase, Cochrane Library, and Web of Science databases from January 2000 to November 2024 for completed studies. We reviewed 30,976 papers, and 27 met inclusion criteria, which were as follows: RCTs published in English, RCTs that involved ACL reconstruction with different grafts (autografts, allografts, and artificial ligaments) with fully described surgical techniques, RCTs that reported on at least one of the relevant outcome measures (preoperative and post-operative IKDC objective score, Lysholm score, Tegner score, Lachman test, pivot-shift test, KT-1000/2000 arthrometer side-to-side difference, and risk of graft reinjury), and RCTs that reported a minimum follow-up period of 12 months with at least 80% follow-up completeness. The Cochrane risk-of-bias tool (RoB 2.0) was used to assess the quality of the included studies, with 18 studies judged as being at a low risk of bias and 9 studies assessed as being at an unclear risk of bias. In total, 2572 patients were treated with 16 different types of grafts. The mean length of follow-up was 47 months. The mean age of the included patients was 28 years, and 71% of the patients were male. A Bayesian network meta-analysis was performed via the “gemtc” and “coda” packages of R Studio, version 4.2.1, and the “networkplot” command of Stata 15

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software was used to plot network relationships for direct and indirect comparisons between different interventions. The surface under the cumulative ranking curve (SUCRA) was calculated to rank the intervention effects of different interventions. A SUCRA value approaching 1 (100%) indicates a greater probability that the corresponding treatment is superior in terms of efficacy or safety. The most common treatment comparisons were the single-bundle 4-strand semitendinosus with gracilis tendon autograft versus the bone–patellar tendon–bone autograft and the single-bundle 4-strand semitendinosus with gracilis tendon autograft versus the double-bundle 4-strand semitendinosus with gracilis tendon autograft.

**Results** Although there were no differences when comparing different grafts for traditional pairwise comparisons of patient-reported outcomes, joint stability, and graft reinjury, the network meta-analysis allows ranking of interventions by efficacy or safety through the SUCRA. In terms of patient-reported outcomes, the double-bundle 4-strand semitendinosus with gracilis tendon autograft had the highest ranking for the IKDC objective score (SUCRA 77%), the single-bundle 3-strand semitendinosus tendon autograft was ranked most favorably for the Lysholm score (SUCRA 78%), and the single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament had the highest ranking for the Tegner score (SUCRA 90%). Concerning knee stability, the double-bundle 4-strand semitendinosus with gracilis tendon autograft ranked first (SUCRA 58%) on the Lachman test, the single-bundle 5-strand semitendinosus with gracilis tendon autograft ranked first (SUCRA 64%) in terms of the pivot-shift test, and the bone–patellar tendon–bone autograft was ranked most favorably for the KT-1000/2000 arthrometer side-to-side difference (SUCRA 91%). The most effective treatment in terms of graft reinjury was the double-bundle 4-strand semitendinosus with gracilis tendon autograft (SUCRA 70%). Again, a higher SUCRA value (close to 1) represents higher probability of a treatment being ranked among the most effective interventions.

**Conclusion** In ACL reconstruction, based on the results of SUCRA rankings, patients should be counseled that the double-bundle 4-strand semitendinosus with gracilis tendon autograft and the bone–patellar tendon–bone autograft represent potentially superior surgical techniques for simultaneously enhancing patient comfort, restoring knee anterior stability, and having low reinjury risk, and the single-bundle 5-strand semitendinosus with gracilis tendon autograft has the potential advantage in restoring rotational knee stability. To achieve best patient-reported outcomes, the single-bundle 3/4-strand semitendinosus tendon autograft and the single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament are also recommended. Further comparisons of different grafts should be performed to determine whether

meaningful differences exist between men and women in terms of these surgical outcomes.

*Level of Evidence* Level I, therapeutic study.

## Introduction

An ACL injury is one of the most common sports injuries of the knee, with > 120,000 surgical procedures performed to treat it each year in the United States [23, 27]. An ACL injury may result in recurrent knee instability, decreased activity, meniscal injuries, and increased risk of knee osteoarthritis [18]. ACL reconstruction is widely used for patients who experience severe subjective instability during specific activities and who wish to return to certain sports [12, 18, 33]. In clinical practice, many operative methods, including different autografts, allografts, and artificial ligaments, have been applied for ACL reconstruction, but each of these approaches possesses its own set of advantages and disadvantages [5, 8, 10, 13, 22, 35, 39, 40, 47, 48].

Despite the numerous RCTs and meta-analyses that have been conducted to establish the superiority of one graft over another, there is still no consensus on which approach leads to the best postoperative outcomes [3, 6, 9, 11, 17, 26, 37, 55]. Furthermore, the large number of different grafts makes it impossible to compare all treatments. A network meta-analysis allows for the comparison of multiple treatments in the same study, taking into account all existing direct and indirect evidence and potentially ranking them, even if they have not previously been compared in head-to-head RCTs [7]. Although several previous network meta-analyses have been conducted on this topic, none have adequately differentiated the distinct clinical impact of different bundles and strands of grafts [29, 51].

We therefore performed an updated network meta-analysis of RCTs to answer the following questions: Considering different grafts applied in ACL reconstruction, which graft is superior in terms of (1) patient-reported outcomes (International Knee Documentation Committee [IKDC] objective score, Lysholm score, and Tegner score), (2) knee stability results (Lachman test, pivot-shift test, and KT-1000/2000 arthrometer side-to-side difference), and (3) risk of graft reinjury?

## Materials and Methods

This systematic review and network meta-analysis was performed and reported in line with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) Statement [32]. The protocol was prospectively registered with a prespecified methodology on PROSPERO (CRD42024601097).

### Search Strategy and Eligibility Criteria

The inclusion criteria for this study were as follows: (1) RCTs published in English, (2) ACL reconstruction using a graft or artificial ligament (autografts, allografts, and artificial ligaments) with the surgical techniques fully described, (3) at least one of the relevant outcome measures reported (preoperative and postoperative IKDC objective score, Lysholm score, Tegner score, Lachman test, pivot-shift test, KT-1000/2000 arthrometer side-to-side difference, and risk of graft reinjury), and (4) minimum follow-up period of 12 months and at least 80% follow-up completeness at the minimum follow-up time.

The exclusion criteria were as follows: (1) non-RCTs (cohort studies, case-control studies, case reports, or review articles), (2) duplicate publications, (3) full text that could not be obtained, (4) a follow-up period of < 12 months or a follow-up completeness of < 80% at the minimum follow-up time, (5) studies that did not directly compare the outcomes of different grafts, and (6) studies in which patients had fractures on the same side or previous knee surgery (except diagnostic arthroscopy or meniscus surgery).

### Information Sources and Search Strategy

Five authors (CFC, SYQ, TL, LY, YX) independently and systematically searched four databases (PubMed, Embase, Cochrane Library, and Web of Science) to identify studies published from January 2000 to November 2024. The base terms used included “ACL,” “transplants,” and “graft” (Appendices 1-4; <http://links.lww.com/CORR/B432>).

### Study Selection

We imported all the retrieved studies into Endnote and then excluded duplicates. After screening the title or abstract, we filtered out preliminary studies, downloaded the full text, read the full text, and screened out the original studies that fit this systematic review. According to the inclusion and exclusion criteria, three authors (CFC, SYQ, LW) independently selected all the articles and cross-checked them by reviewing the full texts. Any disagreements at the inclusion stage were resolved by discussion with the corresponding author (JL).

### Data Extraction

Data were extracted via a standardized extraction spreadsheet that included title, first author, year of publication, study type, author country, treatment method, and

outcomes. The outcomes evaluated included the preoperative and postoperative IKDC objective score, Lysholm score, Tegner score, Lachman test, pivot-shift test, KT-1000/2000 arthrometer side-to-side difference, and frequency of graft reinjury.

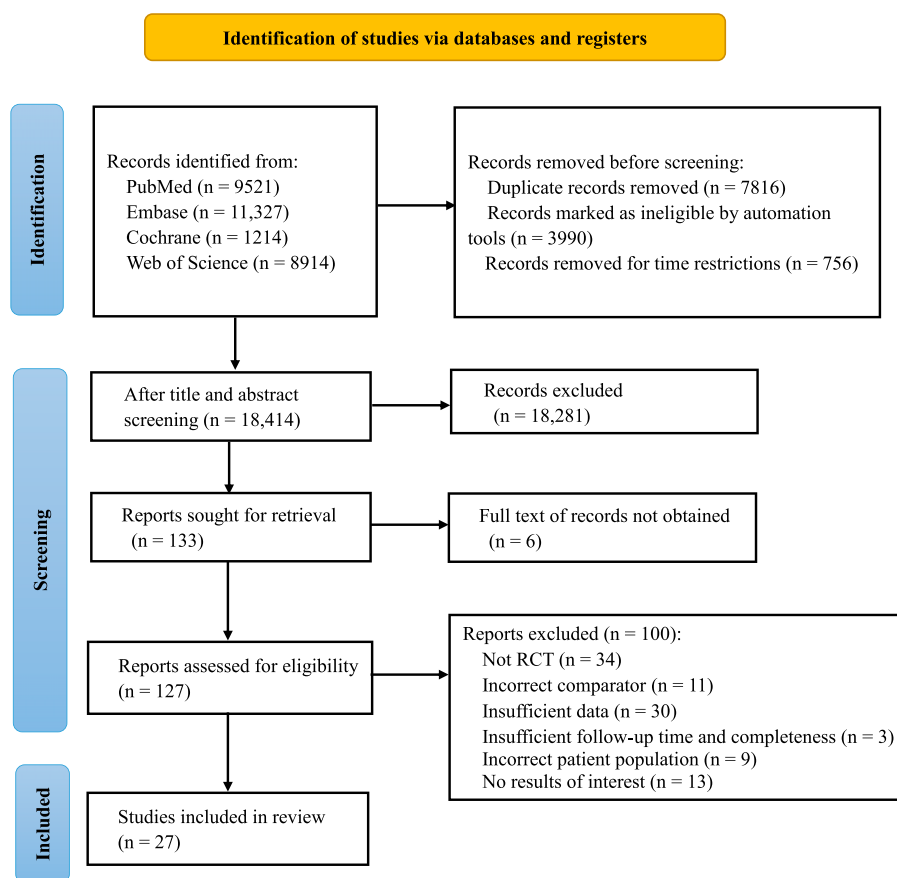
### Study Characteristics

A total of 30,976 records were obtained by searching keywords, including PubMed (n = 9521), Embase (n = 11,327), the Cochrane Library (n = 1214), and the Web of Science (n = 8914). A total of 127 studies underwent full-text review, of which 27 studies [1, 2, 4, 14-16, 19-21, 24, 25, 28, 30, 31, 34, 36, 38, 41-43, 45, 46, 49, 50, 52-54] were ultimately included in the final analysis (Fig. 1).

The 27 studies included in this review were published in English and included a total of 2572 patients (Table 1). Most papers were two-arm studies (93% [25 of 27]), with only two studies (7% [2 of 27]) [25, 28] having three arms. The mean length of follow-up was 47 months. The mean age of the combined patients was 28 years, and 71% of the patients were male. Given that most studies reported that there were no differences in gender composition between the experimental group and the control group, and only four studies [15, 20, 38, 43] analyzed postoperative indicators with gender differences as a subgroup, it was not feasible to include gender differences analysis in this network meta-analysis. The IKDC objective score was evaluated in 14 studies with 1342 patients (Fig. 2A), the Lysholm score was evaluated in 9 studies with a total of 885 patients (Fig. 2B), and the Tegner score was evaluated in 7 studies with a total of 964 patients (Fig. 2C). The Lachman test was evaluated in 16 studies with 1446 patients (Fig. 3A), the pivot-shift test was evaluated in 18 studies with 1710 patients (Fig. 3B), and the KT-1000/2000 side-to-side difference was evaluated in 7 studies with 526 patients (Fig. 3C). Risk of graft reinjury was evaluated in 13 studies with 1210 patients (Fig. 4). The most common treatment comparisons (which contributed to each network analysis) were single-bundle 4-strand semitendinosus with gracilis tendon autograft versus bone-patellar tendon-bone autograft and single-bundle 4-strand semitendinosus with gracilis tendon autograft versus the double-bundle 4-strand semitendinosus with gracilis tendon autograft. A total of 16 different graft types were used for ACL reconstruction in these patients (Supplemental Table 1; <http://links.lww.com/CORR/B432>).

### Risk of Bias and Publication Bias Assessment

Two researchers (CFC, TL) assessed the quality of the included studies with the Cochrane risk-of-bias tool (RoB 2.0). The studies were categorized as “low risk,” “unclear



**Fig. 1** Flow chart of article selection process.

risk,” or “high risk,” and any discrepancies in the assessment were judged after discussion with a third researcher (JL). The reviewers assessed articles for publication bias in five areas: (1) bias arising from the randomization process, (2) bias from deviations from the intended intervention, (3) bias from missing outcome data, (4) bias in outcome measures, and (5) bias in the choice of outcome report. The Network package of Stata 15 was used to draw funnel plots for rough qualitative judgment of publication bias.

In the assessment of the individual trials, 18 studies were judged to be at a low risk of bias, which had low risk of bias in at least four major categories of bias, and 9 studies were judged to be at an unclear risk of bias (Fig. 5A). A cumulative evaluation of the bias revealed that the quality of the studies was overall deemed to be good (Fig. 5B). In the evaluation of the risk of bias across trials, the domains of randomization process, missing outcome data, and selection of the reported results were deemed to be at a low risk of bias. On the contrary, the domains relating to blinding (patients, surgeons, and evaluators) were considered to be at an unclear risk of bias, which was mainly attributed to the nature of surgical interventions. The results of the funnel plots demonstrate that almost all the

points are situated within the funnel and exhibit greater symmetry between the left and right sides, suggesting a low likelihood of publication bias (Supplemental Fig. 1; <http://links.lww.com/CORR/B432>).

### Effect Measures

We used relative risk for dichotomous data and we used mean difference for continuous variable data. Both types of indicators are expressed as 95% confidence intervals. In addition, a  $p$  value  $< 0.05$  denoted statistical significance. A Bayesian network meta-analysis was performed via the “gemtc” and “coda” packages of R Studio, version 4.2.1, and the “networkplot” command of Stata 15 software was used to plot network relationships for direct and indirect comparisons between different interventions. The consistency model was established with an annealing number of 20,000 and an iteration number of 50,000, and the deviation information criterion (DIC) results of the iteration results were further compared with the DIC results of the inconsistency model. If the difference value was  $< 5$ , the data conformed to consistency, and the results of indirect

**Table 1.** Summarized characteristics of included studies

| Author [ref.]       | Year | Country | Type of graft                                                        | Sample size | Age in years | Included outcome | Mean follow-up time in months |
|---------------------|------|---------|----------------------------------------------------------------------|-------------|--------------|------------------|-------------------------------|
| Jansson et al. [20] | 2003 | Finland |                                                                      |             |              |                  |                               |
| Treatment Group 1   |      |         | Bone–patellar tendon–bone autograft                                  | 51          |              | 1, 2, 3, 4, 5    | 24                            |
| Treatment Group 2   |      |         | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 48          |              | 1, 2, 3, 4, 5    | 24                            |
| Gobbi et al. [15]   | 2005 | Italy   |                                                                      |             |              |                  |                               |
| Treatment Group 1   |      |         | Single-bundle 4-strand semitendinosus tendon autograft               | 50          | 31           | 3                | 36                            |
| Treatment Group 2   |      |         | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 47          | 29           | 3                | 36                            |
| Ibrahim et al. [19] | 2005 | Kuwait  |                                                                      |             |              |                  |                               |
| Treatment Group 1   |      |         | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 40          | 22           | 1, 2             | 81                            |
| Treatment Group 2   |      |         | Double-bundle 4-strand semitendinosus with gracilis tendon autograft | 45          | 22           | 1, 2             | 81                            |
| Laxdal et al. [25]  | 2005 | Sweden  |                                                                      |             |              |                  |                               |
| Treatment Group 1   |      |         | Bone–patellar tendon–bone autograft                                  | 40          | 28           | 4, 5, 7          | 26                            |
| Treatment Group 2   |      |         | Single-bundle 3-strand semitendinosus tendon autograft               | 39          | 24           | 4, 5, 7          | 26                            |
| Treatment Group 3   |      |         | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 39          | 26           | 4, 5, 7          | 26                            |
| Järvelä [21]        | 2007 | Finland |                                                                      |             |              |                  |                               |
| Treatment Group 1   |      |         | Double-bundle 4-strand semitendinosus with gracilis tendon autograft | 35          | 33 ± 9.0     | 1, 3, 4, 7       | 14                            |
| Treatment Group 2   |      |         | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 30          | 33 ± 9.0     | 1, 3, 4, 7       | 14                            |
| Siebold et al. [41] | 2008 | Germany |                                                                      |             |              |                  |                               |
| Treatment Group 1   |      |         | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 35          | 28           | 6, 7             | 19                            |
| Treatment Group 2   |      |         | Double-bundle 4-strand semitendinosus with gracilis tendon autograft | 35          | 29           | 6, 7             | 19                            |
| Aglietti et al. [1] | 2010 | Italy   |                                                                      |             |              |                  |                               |
| Treatment Group 1   |      |         | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 35          | 28           | 6, 7             | 24                            |
| Treatment Group 2   |      |         | Double-bundle 4-strand semitendinosus with gracilis tendon autograft | 35          | 28           | 6, 7             | 24                            |

Table 1. continued

| Author [ref.]         | Year | Country  | Type of graft                                                        | Sample size | Age in years | Included outcome | Mean follow-up time in months |
|-----------------------|------|----------|----------------------------------------------------------------------|-------------|--------------|------------------|-------------------------------|
| Barenius et al. [2]   | 2010 | Sweden   |                                                                      |             |              |                  |                               |
| Treatment Group 1     |      |          | Bone–patellar tendon–bone autograft                                  | 78          | 33 ± 6.3     | 3, 4, 5, 7       | 100.8                         |
| Treatment Group 2     |      |          | Single-bundle 4-strand semitendinosus tendon autograft               | 75          | 35 ± 7.5     | 3, 4, 5, 7       | 100.8                         |
| Ghalayini et al. [14] | 2010 | UK       |                                                                      |             |              |                  |                               |
| Treatment Group 1     |      |          | Synthetic artificial ligament                                        | 24          | 31.7 ± 6.7   | 2, 3, 7          | 60                            |
| Treatment Group 2     |      |          | Bone–patellar tendon–bone autograft                                  | 26          | 30.9 ± 6.7   | 2, 3, 7          | 60                            |
| Volpi et al. [49]     | 2010 | Italy    |                                                                      |             |              |                  |                               |
| Treatment Group 1     |      |          | Bone–patellar tendon–bone autograft                                  | 20          | 27 ± 5.7     | 3                | 31                            |
| Treatment Group 2     |      |          | Double-bundle 4-strand semitendinosus with gracilis tendon autograft | 20          | 30 ± 4.9     | 3                | 28                            |
| Sun et al. [45]       | 2011 | China    |                                                                      |             |              |                  |                               |
| Treatment Group 1     |      |          | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 91          | 29.6 ± 6.9   | 1, 2, 3, 4, 5    | 91.2                          |
| Treatment Group 2     |      |          | Single-bundle 4-strand semitendinosus with gracilis tendon allograft | 95          | 31.2 ± 8.3   | 1, 2, 3, 4, 5    | 94.8                          |
| Noh et al. [31]       | 2011 | Korea    |                                                                      |             |              |                  |                               |
| Treatment Group 1     |      |          | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 33          | 23           | 1, 2, 3, 4, 5    | 28.1                          |
| Treatment Group 2     |      |          | Single-bundle 2-strand Achilles tendon allograft                     | 32          | 22           | 1, 2, 3, 4, 5    | 31.6                          |
| Sajovic et al. [36]   | 2011 | Slovenia |                                                                      |             |              |                  |                               |
| Treatment Group 1     |      |          | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 32          | 36           | 1, 2, 7          | 132                           |
| Treatment Group 2     |      |          | Bone–patellar tendon–bone autograft                                  | 32          | 38           | 1, 2, 7          | 132                           |
| Lui et al. [28]       | 2012 | China    |                                                                      |             |              |                  |                               |
| Treatment Group 1     |      |          | Bone–patellar tendon–bone autograft                                  | 14          | 26.5 ± 5.0   | 2                | 12                            |
| Treatment Group 2     |      |          | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 21          | 23.2 ± 3.9   | 2                | 12                            |
| Treatment Group 3     |      |          | Double-bundle 4-strand semitendinosus with gracilis tendon autograft | 17          | 26.4 ± 4.9   | 2                | 12                            |
| Noh et al. [30]       | 2013 | Korea    |                                                                      |             |              |                  |                               |
| Treatment Group 1     |      |          | Single-bundle 2-strand Achilles tendon with bone block allograft     | 33          | 23           | 1, 2, 3, 4, 6    | 30.9                          |



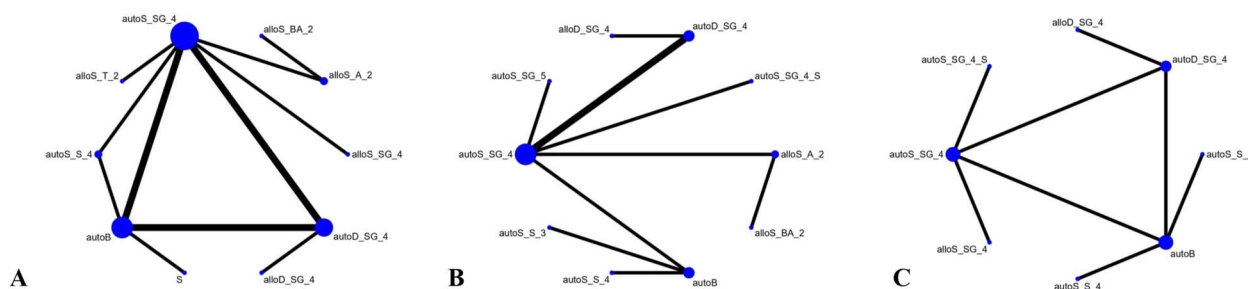
Table 1. continued

| Author [ref.]      | Year | Country | Type of graft                                                        | Sample size | Age in years | Included outcome | Mean follow-up time in months |
|--------------------|------|---------|----------------------------------------------------------------------|-------------|--------------|------------------|-------------------------------|
| Treatment Group 2  | 2013 | Korea   | Single-bundle 2-strand Achilles tendon allograft                     | 34          | 22           | 1, 2, 3, 4, 6    | 30.9                          |
| Song et al. [43]   |      |         | Double-bundle 4-strand tibialis tendon allograft                     | 65          | 30.3         | 1, 2, 7          | 65.6                          |
| Treatment Group 1  |      |         | Single-bundle 2-strand tibialis tendon allograft                     | 65          | 35.5         | 1, 2, 7          | 68.4                          |
| Treatment Group 2  | 2014 | Iran    | Bone–patellar tendon–bone autograft                                  | 46          | 30.8 ± 4.5   | 1, 2, 3          | 36                            |
| Razi et al. [34]   |      |         | Double-bundle 4-strand semitendinosus with gracilis tendon autograft | 41          | 28.2 ± 3.7   | 1, 2, 3          | 36                            |
| Treatment Group 1  |      |         | Double-bundle 4-strand semitendinosus with gracilis tendon autograft | 76          | 28.2 ± 12.6  | 1, 5, 6, 7       | 24                            |
| Treatment Group 2  | 2016 | Japan   | Bone–patellar tendon–bone autograft                                  | 74          | 27.0 ± 11.9  | 1, 5, 6, 7       | 24                            |
| Sasaki et al. [38] |      |         | Double-bundle 4-strand semitendinosus with gracilis tendon autograft | 78          | 30.5 ± 4.9   | 1, 2, 3, 4, 5    | 55.2                          |
| Treatment Group 1  |      |         | Double-bundle 4-strand semitendinosus with gracilis tendon autograft | 79          | 29.9 ± 6.1   | 1, 2, 3, 4, 5    | 55.2                          |
| Tian et al. [46]   | 2017 | China   | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 30          | 34.9 ± 4.3   | 1                | 34.8                          |
| Treatment Group 1  |      |         | Double-bundle 4-strand gracilis tendon autograft                     | 30          | 34.8 ± 4.2   | 1                | 36.2                          |
| Treatment Group 2  |      |         | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 71          | 30           | 1, 2, 3, 7       | 32.8                          |
| Yoo et al. [52]    | 2018 | Korea   | Single-bundle 2-strand tibialis tendon allograft                     | 70          | 24           | 1, 2, 3, 7       | 34.5                          |
| Treatment Group 1  |      |         | Single-bundle 4-strand peroneus longus tendon autograft              | 62          | 29.1 ± 6.5   | 1, 7             | 29.7                          |
| Treatment Group 2  |      |         | Single-bundle 4-strand semitendinosus tendon autograft               | 62          | 27.9 ± 6.7   | 1, 7             | 30.3                          |
| Bi et al. [4]      | 2019 | China   | Single-bundle 4-strand semitendinosus with gracilis tendon autograft | 78          | 27.6 ± 7.3   | 1, 2, 3, 4, 5, 6 | 28.2                          |
| Zhang et al. [54]  |      |         | Single-bundle 4-strand semitendinosus with gracilis tendon autograft |             |              |                  |                               |

Table 1. continued

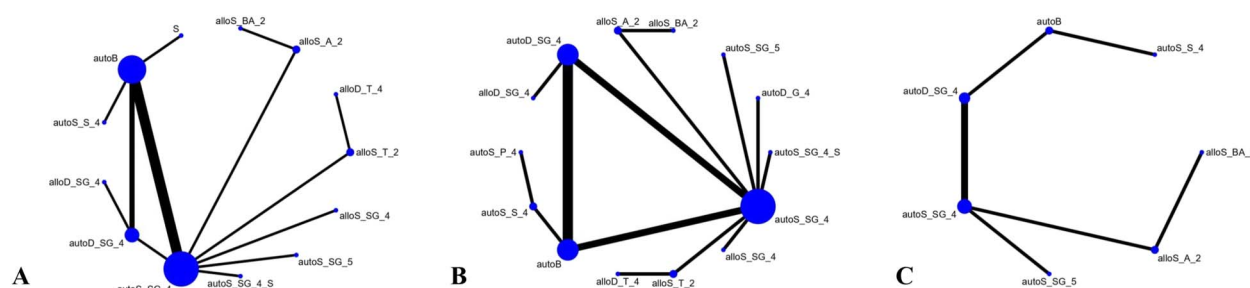
| Author [ref.]            | Year | Country   | Type of graft                                                                                           | Sample size | Age in years | Included outcome | Mean follow-up time in months |
|--------------------------|------|-----------|---------------------------------------------------------------------------------------------------------|-------------|--------------|------------------|-------------------------------|
| Treatment Group 2        |      |           | Double-bundle 4-strand semitendinosus with gracilis tendon autograft                                    | 78          | 25.9 ± 5.2   | 1, 2, 3, 4, 5, 6 | 28.2                          |
| Smith et al. [42]        | 2020 | USA       |                                                                                                         |             |              |                  |                               |
| Treatment Group 1        |      |           | Single-bundle 4-strand semitendinosus tendon autograft                                                  | 32          | 17.6 ± 2.5   | 6, 7             | 24                            |
| Treatment Group 2        |      |           | Bone–patellar tendon–bone autograft                                                                     | 32          | 17.8 ± 2.3   | 6, 7             | 24                            |
| Guglielmetti et al. [16] | 2021 | Brazil    |                                                                                                         |             |              |                  |                               |
| Treatment Group 1        |      |           | Single-bundle 4-strand semitendinosus with gracilis tendon autograft                                    | 31          | 24.6 ± 5.3   | 1, 2, 3, 4, 6    | 24                            |
| Treatment Group 2        |      |           | Bone–patellar tendon–bone autograft                                                                     | 31          | 25.2 ± 5.5   | 1, 2, 3, 4, 6    | 24                            |
| Krishna et al. [24]      | 2021 | Singapore |                                                                                                         |             |              |                  |                               |
| Treatment Group 1        |      |           | Single-bundle 4-strand semitendinosus with gracilis tendon autograft                                    | 28          | 26.3 ± 7.3   | 1, 2, 3, 4, 6    | 24                            |
| Treatment Group 2        |      |           | Single-bundle 5-strand semitendinosus with gracilis tendon autograft                                    | 28          | 27.6 ± 7.3   | 1, 2, 3, 4, 6    | 24                            |
| Zaid et al. [53]         | 2023 | China     |                                                                                                         |             |              |                  |                               |
| Treatment Group 1        |      |           | Single-bundle 4-strand semitendinosus with gracilis tendon autograft                                    | 47          | 26.4 ± 4.0   | 1, 2, 4, 5, 7    | 18                            |
| Treatment Group 2        |      |           | Single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament | 47          | 26.1 ± 3.5   | 1, 2, 4, 5, 7    | 18                            |

Data presented as the number or mean ± SD. 1 = pivot-shift test; 2 = Lachman test; 3 = IKDC objective score; 4 = Lysholm score; 5 = Tegner score; 6 = KT-1000/2000 arthrometer side-to-side difference; 7 = graft reinjury.



**Fig. 2** Network plots for clinical outcomes, including (A) the IKDC objective score, (B) the Lysholm score, and (C) the Tegner score. Abbreviations and full names of the grafts are listed in Supplemental Table 1 (<http://links.lww.com/CORR/B432>).





**Fig. 3** Network plots for knee stability results, including (A) the Lachman test, (B) the pivot-shift test, and (C) the KT-1000/2000 side-to-side difference. Abbreviations and full names of the grafts are listed in Supplemental Table 1 (<http://links.lww.com/CORR/B432>).

comparisons of the network meta-analysis were calculated to be not significantly different from results of the direct comparisons. The surface under the cumulative ranking curve (SUCRA) was calculated to rank the intervention effects of different interventions. A SUCRA value approaching 1 (100%) indicates a greater probability that the corresponding treatment is superior in efficacy or safety rankings.

#### Heterogeneity Assessment

We used the  $I^2$  test to evaluate heterogeneity between studies in both consistency and inconsistency models.  $I^2$  values  $> 50\%$  were deemed as substantial heterogeneity. Conversely,  $I^2$  values  $< 50\%$  were classified as low

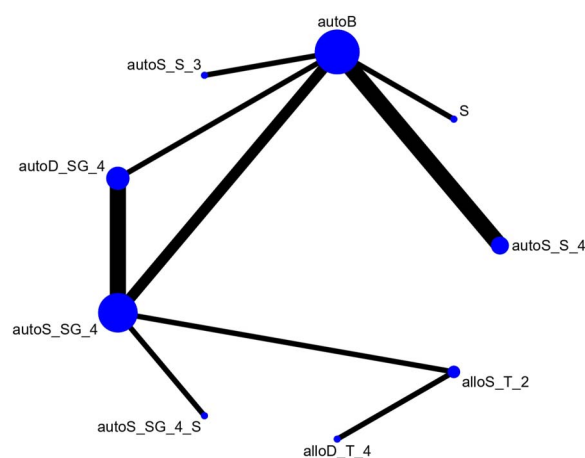
heterogeneity. The heterogeneity assessment results demonstrated that the values were  $< 50\%$  in both consistency and inconsistency models, indicating low heterogeneity in comparisons among studies (Supplemental Table 2; <http://links.lww.com/CORR/B432>). Although heterogeneity among the studies in this network meta-analysis was low, we employed a random-effects model for the analysis because it explicitly accounted for potential minimal heterogeneity.

#### Inconsistency Assessment

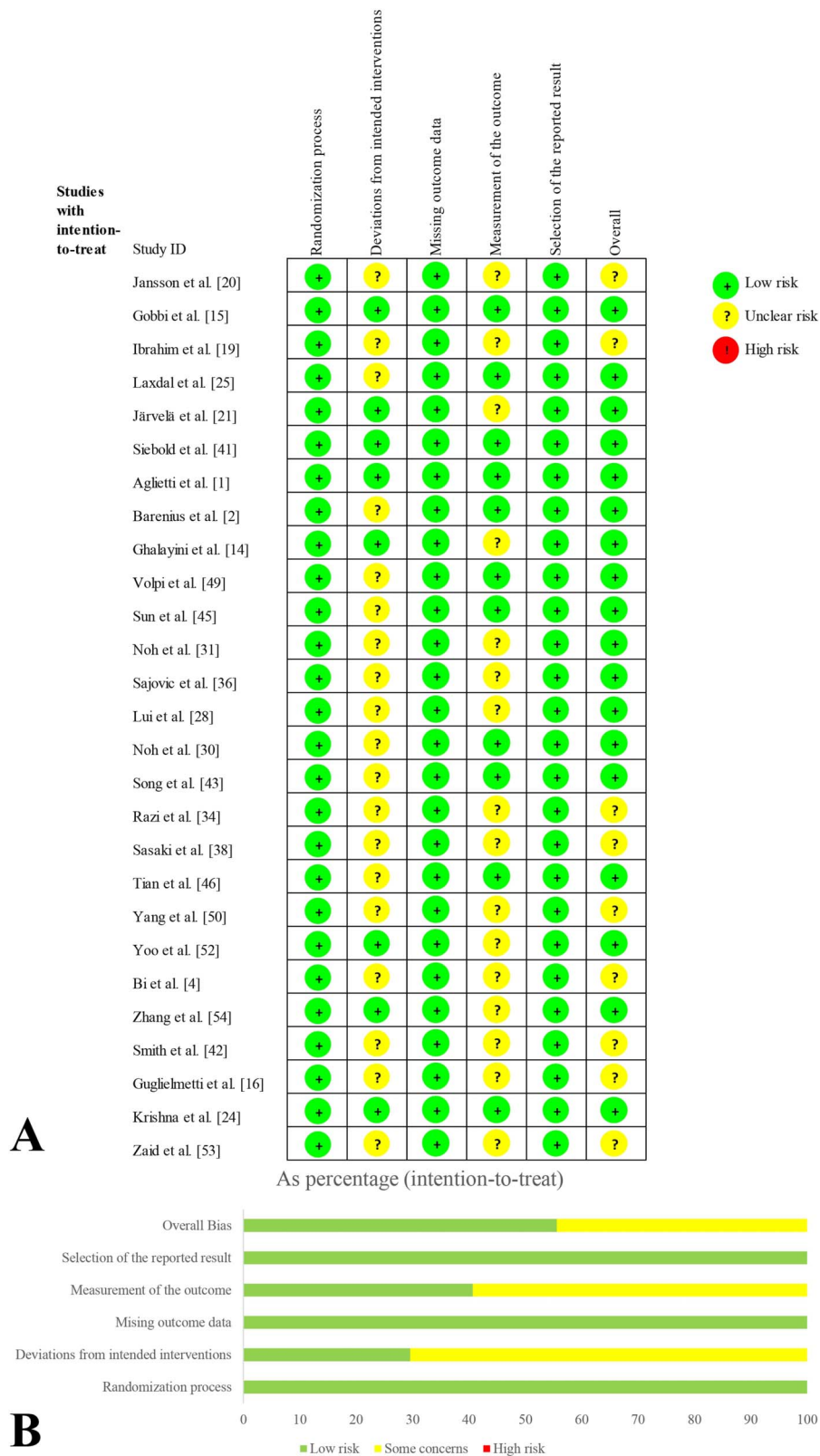
In this study, a local inconsistency test was performed using the node-splitting method for studies with closed loops. The results of local inconsistency analyses revealed no statistically significant deviations in closed-loop comparisons ( $p > 0.05$ ), thus suggesting the validity of the consistency model (Supplemental Figs. 2-6; <http://links.lww.com/CORR/B432>).

#### Sensitivity Assessment

The sensitivity assessment of seven outcomes (IKDC, Lysholm score, Tegner score, Lachman test, pivot-shift test, KT-1000/2000 side-to-side difference, risk of graft reinjury) was conducted via consistency and inconsistency models. The results of posterior mean deviance (Dbar), effective number of parameters (pD), and DIC indicated that the values were similar between the two models (Supplemental Table 2; <http://links.lww.com/CORR/B432>), suggesting no substantial differences in model fit. The results of both inconsistency assessment and sensitivity assessment indicated that the studies included were comparable and that the data were stable, demonstrating consistent effect estimates across alternative model specifications (consistency and inconsistency models), which reflected the good ability



**Fig. 4** Network plots for risk graft reinjury. The straight line denotes directly comparable studies, the thickness of the line segment signifies the quantity of directly comparable studies, the dots indicate the intervention, and the size of the dots represents the sample size for that intervention. Abbreviations and full names of the grafts are listed in Supplemental Table 1 (<http://links.lww.com/CORR/B432>).



**Fig. 5** (A) Risk of bias graph for each study and (B) summary of the risk of bias for the 27 RCTs. A color image accompanies the online version of this article.

of the findings to resist model assumptions and potential uncertainties in the data.

## Results

Although there were no statistical differences when comparing different grafts for traditional pairwise comparisons of patient-reported outcomes, knee joint stability, and risk of graft reinjury (Supplemental Tables 3-9; <http://links.lww.com/CORR/B432>), the network meta-analysis-based SUCRA rankings provides a quantitative basis for ranking the efficacy or safety of different grafts.

### *Patient-reported Outcomes*

The double-bundle 4-strand semitendinosus with gracilis tendon autograft and the bone–patellar tendon–bone autograft were ranked most favorably for the IKDC objective score (SUCRA 77% and 72%, respectively), the single-bundle 3-strand semitendinosus tendon autograft and the single-bundle 4-strand semitendinosus with gracilis tendon autograft were ranked most favorably for the Lysholm score (SUCRA 78% and 68%, respectively), and the single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament and the single-bundle 4-strand semitendinosus with gracilis tendon autograft were ranked most favorably for the Tegner score (SUCRA 90% and 62%, respectively) (Table 2). Again, a higher SUCRA value (close to 1) represents higher probability of a treatment being ranked among the most effective interventions.

### *Knee Stability Results*

With the Lachman test, the double-bundle 4-strand semitendinosus with gracilis tendon autograft ranked first (SUCRA 58%). The single-bundle 5-strand semitendinosus with gracilis tendon autograft and the double-bundle 4-strand gracilis tendon autograft were ranked most favorably (SUCRA 64% and 61%, respectively) in terms of the pivot-shift test. The bone–patellar tendon–bone autograft and the double-bundle 4-strand semitendinosus with gracilis tendon autograft were ranked most favorably for the KT-1000/2000 arthrometer side-to-side difference (SUCRA 91% and 71%, respectively) (Table 3).

### *Risk of Graft Reinjury*

The three most effective treatments in terms of low risk of graft reinjury were the double-bundle 4-strand

semitendinosus with gracilis tendon autograft (SUCRA 70%), the synthetic artificial ligament (64%), and the bone–patellar tendon–bone autograft (61%) (Table 4).

## Discussion

In clinical practice, different methods (autografts, allografts, and artificial ligaments) have been used for ACL reconstruction. Despite an abundance of RCTs and resultant meta-analyses, there is still no consensus on which approach leads to the best postoperative outcomes. Moreover, the substantial number of various grafts makes it unfeasible to compare all treatments in two-arm RCTs or traditional meta-analyses. We therefore conducted a network meta-analysis to compare different grafts for ACL reconstruction, as many had not previously been compared in head-to-head RCTs. This was achieved by pooling all study data and summarizing results of both direct and indirect comparisons, along with the rank of the included treatments. While traditional pairwise comparisons (similar to traditional meta-analyses) showed no differences in postoperative outcomes among grafts, the network meta-analysis enables ranking of therapeutic interventions by efficacy or safety using the SUCRA, which is a key advantage of this methodology. We found that the double-bundle 4-strand semitendinosus with gracilis tendon autograft had the highest probability of being superior in terms of the IKDC objective score, Lachman test, and risk of graft reinjury; the single-bundle 3-strand semitendinosus tendon autograft and the single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament were ranked most favorably for the Lysholm and Tegner scores, respectively; the single-bundle 5-strand semitendinosus with gracilis tendon autograft and the bone–patellar tendon–bone autograft ranked first in terms of the pivot-shift test and the KT-1000/2000 arthrometer side-to-side difference, respectively. Notably, SUCRA rankings should be interpreted as probable trends rather than definitive conclusions. Clinicians should balance SUCRA rankings with direct pairwise comparisons where those are available. For example, if two grafts have similar SUCRA values and no differences in direct comparisons, the choice may reasonably depend on surgeon experience or the patient's subjective preferences rather than model-derived probabilities alone.

### *Limitations*

This network meta-analysis had several limitations. First, given that most studies merely reported that there was no statistically significant difference in gender composition between the experimental group and the control group, and

**Table 2.** Summarized results of SUCRA values for patient-reported outcomes in the included studies

| Rank | Patient-reported outcome                                                   |                                                                                                               |                                                                                                               |
|------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|      | IKDC objective score                                                       | Lysholm score                                                                                                 | Tegner score                                                                                                  |
| 1    | Double-bundle 4-strand semitendinosus with gracilis tendon autograft (77%) | Single-bundle 3-strand semitendinosus tendon autograft (78%)                                                  | Single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament (90%) |
| 2    | Bone–patellar tendon–bone autograft (72%)                                  | Single-bundle 4-strand semitendinosus with gracilis tendon autograft (68%)                                    | Single-bundle 4-strand semitendinosus with gracilis tendon autograft (62%)                                    |
| 3    | Single-bundle 4-strand semitendinosus tendon autograft (67%)               | Single-bundle 5-strand semitendinosus with gracilis tendon autograft (66%)                                    | Single-bundle 3-strand semitendinosus tendon autograft (57%)                                                  |
| 4    | Single-bundle 2-strand Achilles tendon allograft (65%)                     | Single-bundle 2-strand Achilles tendon allograft (55%)                                                        | Bone–patellar tendon–bone autograft (53%)                                                                     |
| 5    | Single-bundle 4-strand semitendinosus with gracilis tendon autograft (55%) | Single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament (55%) | Single-bundle 4-strand semitendinosus with gracilis tendon allograft (49%)                                    |
| 6    | Single-bundle 2-strand tibialis tendon allograft (49%)                     | Single-bundle 2-strand Achilles tendon with bone block allograft (45%)                                        | Double-bundle 4-strand semitendinosus with gracilis tendon autograft (44%)                                    |
| 7    | Single-bundle 4-strand semitendinosus with gracilis tendon allograft (46%) | Single-bundle 4-strand semitendinosus tendon autograft (41%)                                                  | Double-bundle 4-strand semitendinosus with gracilis tendon allograft (35%)                                    |
| 8    | Double-bundle 4-strand semitendinosus with gracilis tendon allograft (31%) | Bone–patellar tendon–bone autograft (39%)                                                                     | Single-bundle 4-strand semitendinosus tendon autograft (12%)                                                  |
| 9    | Single-bundle 2-strand Achilles tendon with bone block allograft (26%)     | Double-bundle 4-strand semitendinosus with gracilis tendon allograft (28%)                                    | NA                                                                                                            |
| 10   | Synthetic artificial ligament (12%)                                        | Double-bundle 4-strand semitendinosus with gracilis tendon autograft (25%)                                    | NA                                                                                                            |

A higher SUCRA value (approaching 100%) indicates a greater probability that the corresponding treatment is superior in terms of efficacy or safety. In other words, a higher SUCRA value indicates a better ranking. For example, double-bundle 4-strand semitendinosus with gracilis tendon autograft has a highest SUCRA of 77% in IKDC objective score, which means that it has a 77% probability of being the best treatment for the IKDC objective score (Rank 1); bone–patellar tendon–bone autograft has a SUCRA of 72%, positioning it as the second best treatment in terms of the IKDC objective score (Rank 2). NA = not available.

only four studies [15, 20, 38, 43] analyzed postoperative indicators with gender differences as a subgroup, it was not feasible to include gender differences analysis in this network meta-analysis. Thus, the reader should be cautious when applying the results of this study to both genders. Second, we acknowledge that a proportion of the included studies were at varying degrees of unclear risk of bias. The domain related to blinding was identified as having an unclear risk of bias, primarily because of the difficulty of blinding surgeons, which is a methodological constraint common to surgical interventions. It is important to note that most of the assessments of postoperative outcome assessments were conducted by nonsurgical clinicians or

rehabilitation specialists, thereby mitigating potential bias introduced by surgeon unblinding. However, because of awareness of treatment allocation, some assessors' evaluation of subjective endpoints, such as Lachman tests and pivot-shift tests, may have potential risk. This highlights the need for caution in interpreting findings in clinical applications. Third, inconsistencies in postoperative rehabilitation protocols and follow-up time may have introduced confounding factors into this study. Nevertheless, the minimum follow-up time of all studies was > 12 months, the average follow-up time was 47 months, and all patients underwent a series of standardized rehabilitation exercises according to different rehabilitation

**Table 3.** Summarized results of SUCRA values for knee stability results in the included studies

| Rank | Knee stability results                                                                                        |                                                                                                               |                                                                            |
|------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|      | Lachman test                                                                                                  | Pivot-shift test                                                                                              | SSD                                                                        |
| 1    | Double-bundle 4-strand semitendinosus with gracilis tendon autograft (58%)                                    | Single-bundle 5-strand semitendinosus with gracilis tendon autograft (64%)                                    | Bone–patellar tendon–bone autograft (91%)                                  |
| 2    | Double-bundle 4-strand semitendinosus with gracilis tendon allograft (56%)                                    | Double-bundle 4-strand with gracilis tendon autograft (61%)                                                   | Double-bundle 4-strand semitendinosus with gracilis tendon autograft (71%) |
| 3    | Single-bundle 4-strand semitendinosus with gracilis tendon autograft (56%)                                    | Single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament (58%) | Single-bundle 4-strand semitendinosus tendon autograft (59%)               |
| 4    | Single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament (54%) | Single-bundle 4-strand semitendinosus with gracilis tendon autograft (58%)                                    | Single-bundle 5-strand semitendinosus with gracilis tendon autograft (46%) |
| 5    | Double-bundle 4-strand tibialis tendon allograft (54%)                                                        | Single-bundle 4-strand semitendinosus with gracilis tendon allograft (55%)                                    | Single-bundle 4-strand semitendinosus with gracilis tendon autograft (33%) |
| 6    | Bone–patellar tendon–bone autograft (55%)                                                                     | Single-bundle 2-strand Achilles tendon allograft (55%)                                                        | Single-bundle 2-strand Achilles tendon allograft (27%)                     |
| 7    | Single-bundle 4-strand semitendinosus tendon autograft (53%)                                                  | Single-bundle 2-strand Achilles tendon with bone block allograft (54%)                                        | Single-bundle 2-strand Achilles tendon with bone block allograft (23%)     |
| 8    | Single-bundle 4-strand semitendinosus with gracilis tendon allograft (53%)                                    | Bone–patellar tendon–bone autograft (51%)                                                                     | NA                                                                         |
| 9    | Single-bundle 2-strand tibialis tendon allograft (51%)                                                        | Double-bundle 4-strand semitendinosus with gracilis tendon allograft (44%)                                    | NA                                                                         |
| 10   | Synthetic artificial ligament (51%)                                                                           | Double-bundle 4-strand tibialis tendon allograft (42%)                                                        | NA                                                                         |
| 11   | Single-bundle 5-strand semitendinosus with gracilis tendon autograft (50%)                                    | Double-bundle 4-strand semitendinosus with gracilis tendon autograft (42%)                                    | NA                                                                         |
| 12   | Single-bundle 2-strand Achilles tendon allograft (34%)                                                        | Single-bundle 4-strand peroneus longus tendon autograft (41%)                                                 | NA                                                                         |
| 13   | Single-bundle 2-strand Achilles tendon with bone block allograft (30%)                                        | Single-bundle 4-strand semitendinosus tendon autograft (38%)                                                  | NA                                                                         |
| 14   | NA                                                                                                            | Single-bundle 2-strand tibialis tendon allograft (37%)                                                        | NA                                                                         |

A higher SUCRA value (approaching 100%) indicates a greater probability that the corresponding treatment is superior in terms of efficacy or safety. In other words, a higher SUCRA value indicates a better ranking. For example, double-bundle 4-strand semitendinosus with gracilis tendon autograft has a highest SUCRA of 58% in the Lachman test, which means that it has a 58% probability of being the best treatment for the Lachman test (Rank 1); double-bundle 4-strand semitendinosus with gracilis tendon allograft has a SUCRA of 56%, positioning it as the second best treatment in terms of the Lachman test (Rank 2). SSD = KT-1000/2000 arthrometer side-to-side difference.

programs. Therefore, this limitation does not seem disqualifying to us. To improve the quality of meta-analyses, we recommend that future studies include high-quality

RCTs with similar follow-up periods and homogeneous rehabilitation programs. Finally, there were no statistical differences in traditional pairwise comparisons for all



**Table 4.** Summarized results of SUCRA values for risk of graft reinjury in the included studies

| Rank | Risk of graft reinjury                                                                                        |
|------|---------------------------------------------------------------------------------------------------------------|
| 1    | Double-bundle 4-strand semitendinosus with gracilis tendon autograft (70%)                                    |
| 2    | Synthetic artificial ligament (64%)                                                                           |
| 3    | Bone–patellar tendon–bone autograft (61%)                                                                     |
| 4    | Single-bundle 2-strand tibialis tendon allograft (50%)                                                        |
| 5    | Single-bundle 4-strand semitendinosus with gracilis tendon autograft (48%)                                    |
| 6    | Single-bundle 4-strand semitendinosus tendon autograft (42%)                                                  |
| 7    | Single-bundle 3-strand semitendinosus tendon autograft (40%)                                                  |
| 8    | Single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament (39%) |
| 9    | Double-bundle 4-strand tibialis tendon allograft (36%)                                                        |

A higher SUCRA value (approaching 100%) indicates a greater probability that the corresponding treatment is superior in terms of efficacy or safety. In other words, a higher SUCRA value indicates a better ranking. For example, double-bundle 4-strand semitendinosus with gracilis tendon autograft has a highest SUCRA of 70% in risk of graft reinjury, which means that it has a 70% probability of being the best treatment for the risk of graft reinjury (Rank 1); synthetic artificial ligament has a SUCRA of 64%, positioning it as the second best treatment in terms of the risk of graft reinjury (Rank 2).

postoperative outcomes among various grafts, and the ranking results of SUCRA were based on point estimate probabilities. As a consequence, it is important to consider higher ranking results as potential superiority rather than definitive evidence.

#### Patient-reported Outcomes

We found no difference in patient-reported outcomes among different grafts. However, the SUCRA ranking was performed by aggregation of nondirect evidence in our study, and the results showed that the double-bundle 4-strand semitendinosus with gracilis tendon autograft, the single-bundle 3/4-strand semitendinosus tendon autograft, and the single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament represent potentially superior

surgical techniques in terms of the IKDC objective score, the Lysholm score, and the Tegner score, respectively. Therefore, we recommend that the grafts mentioned above be potentially considered for all patients concerned about postoperative comfort. Because the current postoperative assessment scale lacks an assessment of the patient's postoperative proprioceptive recovery, we suggest that attention should be paid to the inclusion of this indicator in the future, as it is one of the most important factors in the maintenance of the dynamic stability of the knee joint.

#### Knee Stability Results

This network meta-analysis revealed no difference in knee stability results among various grafts. In contrast to our findings, Sun et al. [44] found that ACL reconstruction with synthetic ligaments achieved better instrumented laxity than autografts. The difference may be a function of the use of two different autografts (bone–patellar tendon–bone graft and 4-strand hamstring tendon graft) in their study to compare with synthetic ligaments, which introduced a confounding factor into the analysis. Based on SUCRA ranking, the double-bundle 4-strand semitendinosus with gracilis tendon autograft and the bone–patellar tendon–bone autograft may reasonably be recommended for individuals who regularly participate in sharp cutting exercises, as evidenced by the SUCRA ranking, which indicates that these grafts are the most likely to be advantageous in restoring anterior stability to patients' knees.

#### Risk of Graft Reinjury

Overall, risk of graft reinjury did not differ across all grafts. Based on the SUCRA ranking results, three graft types (the double-bundle 4-strand semitendinosus with gracilis tendon autograft, the synthetic artificial ligament, and the bone–patellar tendon–bone autograft) are potentially recommended for primary ACL reconstruction in terms of low graft reinjury risk. The difference in reinjury risk should be considered as a component in a broader conversation with each patient about graft selection, which should also include potential differences in rehabilitation time, donor site morbidity, recovery of proprioception, and other complications. Moreover, further studies should focus on whether meaningful differences mentioned above are present in both men and women.

#### Conclusion

In ACL reconstruction, based on the results of SUCRA rankings, patients should be counseled that the double-bundle 4-strand semitendinosus with gracilis tendon



autograft and the bone–patellar tendon–bone autograft represent potentially superior surgical techniques for simultaneously enhancing patient comfort, restoring knee anterior stability, and minimizing reinjury risk, and the single-bundle 5-strand semitendinosus with gracilis tendon autograft has the potential advantage in restoring rotational knee stability. To achieve best patient-reported outcomes, the single-bundle 3/4-strand semitendinosus tendon autograft and the single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament are also recommended for clinicians. Considering these findings, comparisons of different grafts should be further conducted to determine whether meaningful differences exist between men and women in terms of these surgical outcomes.

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